

Research paper

MutVis: Automated framework for analysis and visualization of mutational signatures in pathogenic bacterial strains

Akshatha Prasanna, Vidya Niranjana^{*}

Department of Biotechnology, RV College of Engineering, Bengaluru, Karnataka 560059, India

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ABSTRACT

In recent years, mutational signature analysis has become a routine practice in cancer genomics for classification and diagnosis. Characterizing mutational signatures across species or within genomes of a bacteria helps in understanding their evolution and adaptation. However, an integrated framework for analysis and visualization of mutational signatures in bacterial genome is lacking. Hence, we aim to develop an integrated, automated, open-source and user-friendly framework called MutVis to analyze mutational signatures from bacterial whole genome next generation sequencing data. The current framework integrates various publicly available packages using Snakemake workflow management software, Python and R scripting. MutVis supports variant calling, transition (Ti) and transversion (Tv) graphical representation, generation of mutational count matrix, graphical visualization of base-pair substitution spectrum (BPSs) and mutation signatures extraction. TvTi plots provide the 6 base substitution classification for both genome and gene level. Further resolution of base pair substitution classification is provided as 96-profile BPSs plot. Mutation signatures is derived based on the characteristic pattern observed in BPSs using non-negative matrix factorization. Relative contribution of signatures is given as hierarchically clustered heatmap. This provides information on active signatures in the individual given sample and classify samples according to signature contributions. We demonstrated the MutVis framework using geographically different strains of *Mycobacterium tuberculosis*, downloaded from PATRIC TB-ARC Antibiotic Resistance Catalog ($n = 963$). The current framework can be used to study mutation biases and characteristic mutational signatures in bacterial genomes and is freely available at <https://github.com/AkshathaPrasanna/MutVis>.

1. Introduction

Evolution of a bacteria is driven by mutations which are selected forward under environmental and immune pressures (Hersberg, 2015). In context of antibiotic resistance, these mutations will help in the survival of a bacteria against antibiotics and emerge as a drug resistant strain (Woodford and Ellington, 2007). This process is entirely not random and follows a specific pattern. This specific pattern of mutations is called as mutation signature (Alexandrov and Stratton, 2014). Mutation signature analysis can assist in identifying the series of mutations in the bacterial genome and its association with the external mutagens causing them. This helps to classify the resistant type, virulent type or geographic type of the bacterial strains.

Recently, several packages are available for generating mutational signatures such as - deconstructSigs (Rosenthal et al., 2016), mafTools (Mayakonda et al., 2018), SomaticSignatures (Gehring et al., 2015),

SigProfilerMatrixGenerator (Bergstrom et al., 2019), signeR (Rosales et al., 2017), and MutationalPattern (Blokzijl et al., 2018). However, these tools greatly focus on the study of cancer genomes and has significantly helped to advance the understanding of mutagenesis and the development of cancer diagnosis (Maura et al., 2019; Wang et al., 2013; Kim et al., 2019; Robinson et al., 2017). The required inputs for the existing mutational signature analysis packages are Single Nucleotide Polymorphisms (SNPs) and reference genome. By comparing the information of SNPs with the reference genome, the mutational signatures are extracted.

Since the evolution in bacteria is caused by mutation, exploring mutational signature analysis approach will enhance our understanding of the mutation pattern in bacterial genome. This has a large potential utility in clinical and diagnostic microbiology. Next generation Whole Genome Sequencing (WGS) data analysis has proven to be a powerful method for identifying SNPs in bacterial genome (McNerney et al., 2018;

^{*} Corresponding author.

E-mail address: vidya.n@rvce.edu.in (V. Niranjana).

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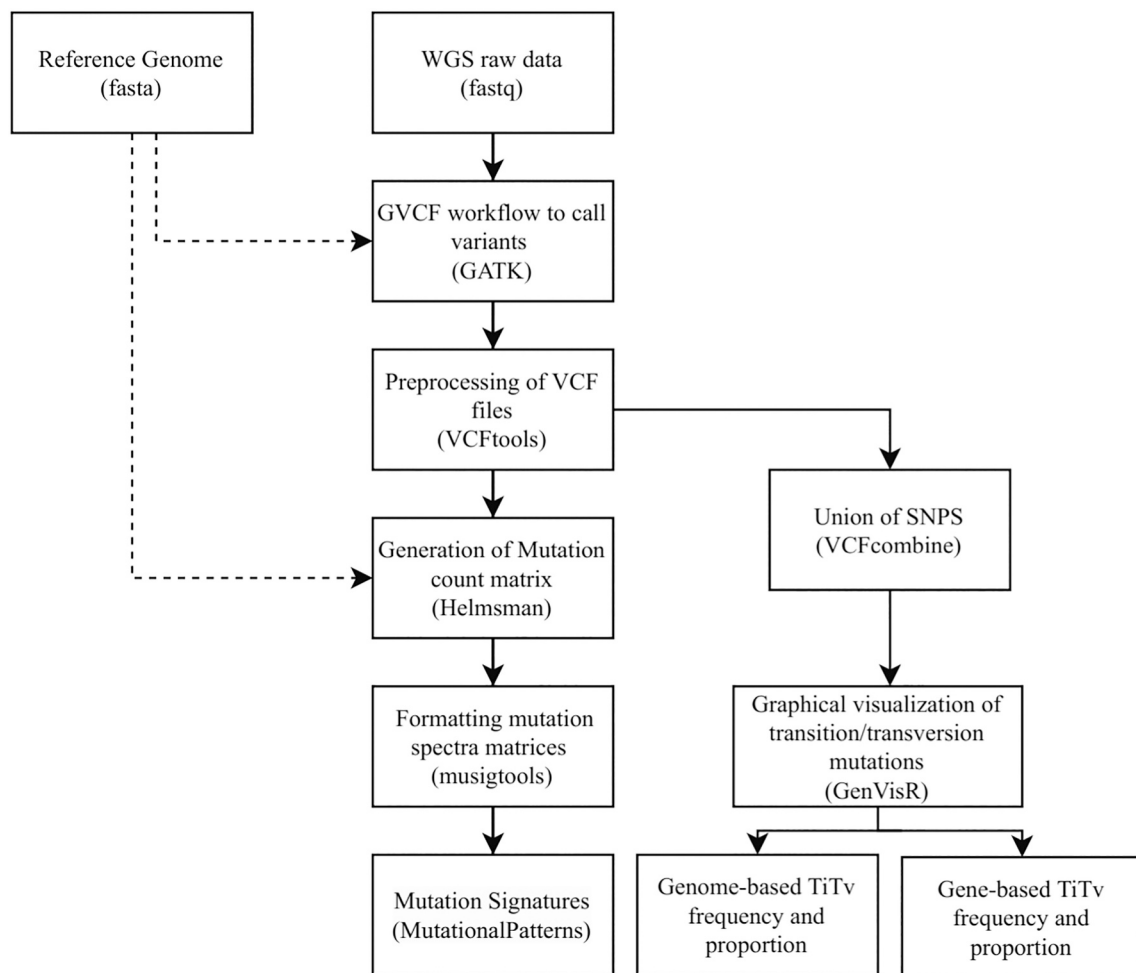


Fig. 1. Schematic representation of MutVis workflow.

Gabrielian et al., 2020; Cooper et al., 2020; Tagini and Greub, 2017).

SNPs comprises of two types – transitions (Ti) and transversions (Tv). Transitions are mutations within the same type of nucleotide: pyrimidine-pyrimidine mutations (C > T) and purine-purine mutations (A > G). Transversions are mutations from a pyrimidine to a purine or vice versa. Mutation signatures is based on these six single base substitutions classes namely G > T or C > A (Tv), G > C or C > G (Tv), G > A or C > T (Ti), A > T or T > A (Tv), A > G or T > C (Ti) and A > C or T > G (Tv) (Skidmore et al., 2016). This is further elaborated to 96 base substitution subclasses by adding the 5' and 3' nucleotide on either side of the mutation to evaluate each substitution class in different sequence contexts (Alexandrov et al., 2013; Alexandrov et al., 2020).

To explore this new methodology of mutational signature analysis in bacterial genomics, we developed a framework called MutVis. The input for the analysis using existing packages is limited to few organism genomes. To incorporate the analysis for all the bacterial genomes, additional tools and formatting of inputs are required. This is addressed in the current framework by utilizing various open source packages and R functions. Among the available packages, MutationalPatterns provides more comprehensive results and flexibility in the integration. Therefore, the MutVis framework is developed based on MutationalPatterns to provide comprehensive analysis of mutation signatures with integration of other open source packages to make it compatible for any bacterial genomes.

The next addressed challenge is the implementation of the presented framework in the clinical setting which demands an easy to use and reproducible workflow without the hurdles of setting up the tools and extensive scripting. To address this, MutVis packs all the necessary tools

required for the critical steps in mutational signature analysis. The workflow can be utilized to install and configure all the analysis tools and their dependencies without the need for manual user intervention. The proposed workflow is automated to handle the multiple data through Snakemake (Köster and Rahmann, 2012) and R scripting. The proposed framework was designed for the analysis of mutational signatures in bacterial genome allowing clinicians or researchers to run through the necessary steps from the raw data to the visualization of results.

2. Methodology

The current framework, MutVis is implemented in Snakemake workflow management software and uses Python and R programming for scripting. It integrates open source tools for construction of transition and transversion plot, mutation count spectra generation and mutation signatures extraction from VCF files. Our framework also supports generation of VCF files for the user provided whole genome sequencing data in FASTQ format.

The Snakemake workflow is defined by a set of rules, with each rule representing an individual step in the analysis. The rule is characterized by a name and includes a set of inputs, outputs, shell commands to invoke command-line tools to generate output. The workflow supports the automatic installation of tools through Conda using BioConda (Grünig et al., 2018), Conda-forge, and default channel repositories in a pre-defined environment. The software stack definition is depicted in an YAML file indicating the source of packages and its versions to be installed. The user can update the versions of the software according to

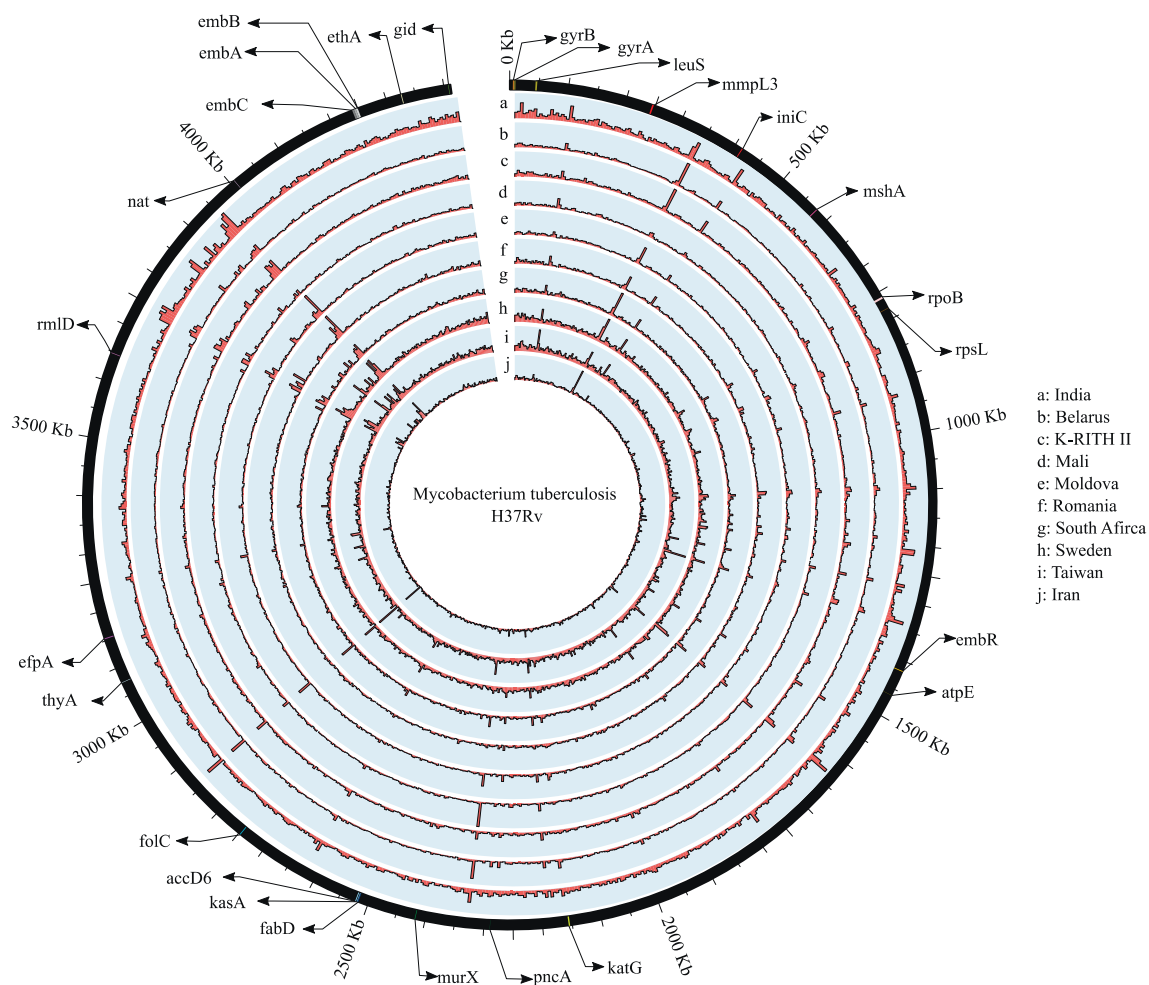


Fig. 2. SNP density plot constructed using Circos.

Note: The outer circle defines the *Mycobacterium tuberculosis* H37Rv genome with gene bands of 26 selected drug resistant genes. Inner circles show the histogram plot of genome-wide SNP density for all the countries, calculated using a sliding-window size of 5 kb.

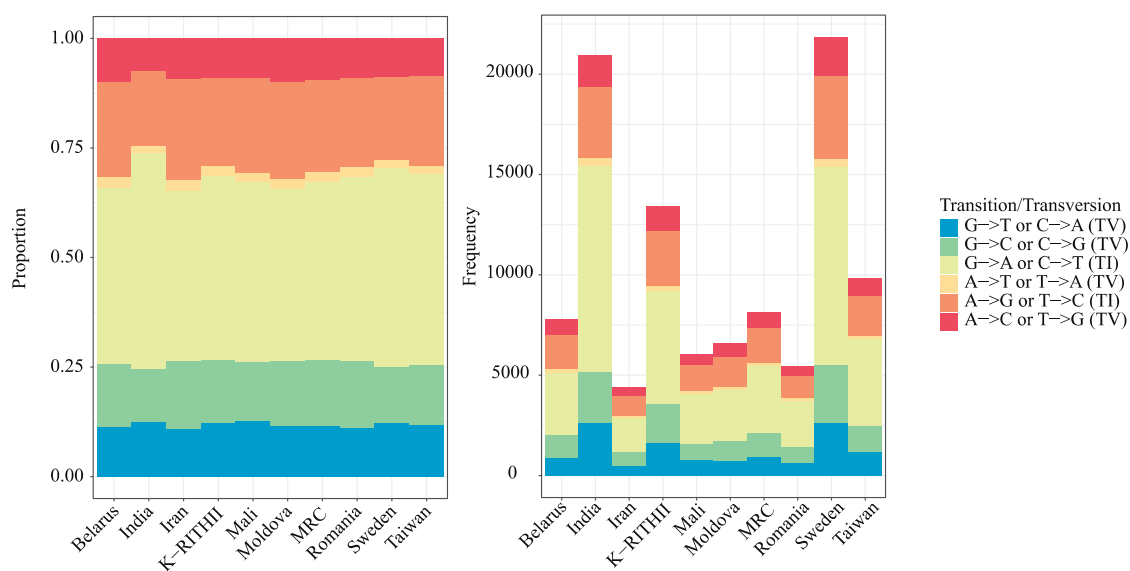


Fig. 3. Proportion and frequency of each transition and transversion mutation types across different geographic types.

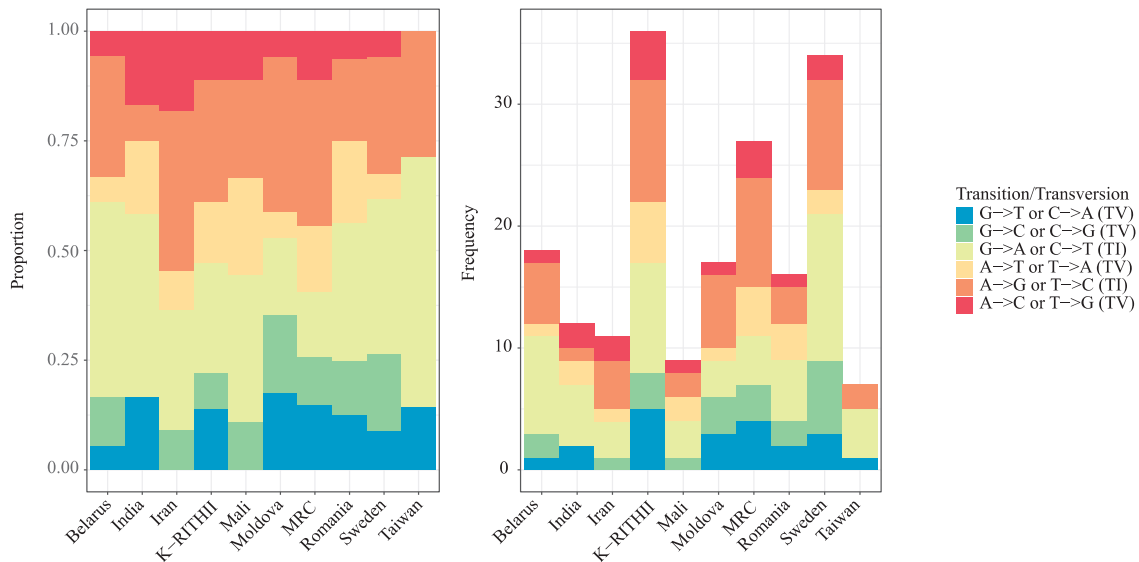


Fig. 4. Proportion and frequency of transition and transversion for the *rpoB* gene across different geographic types.

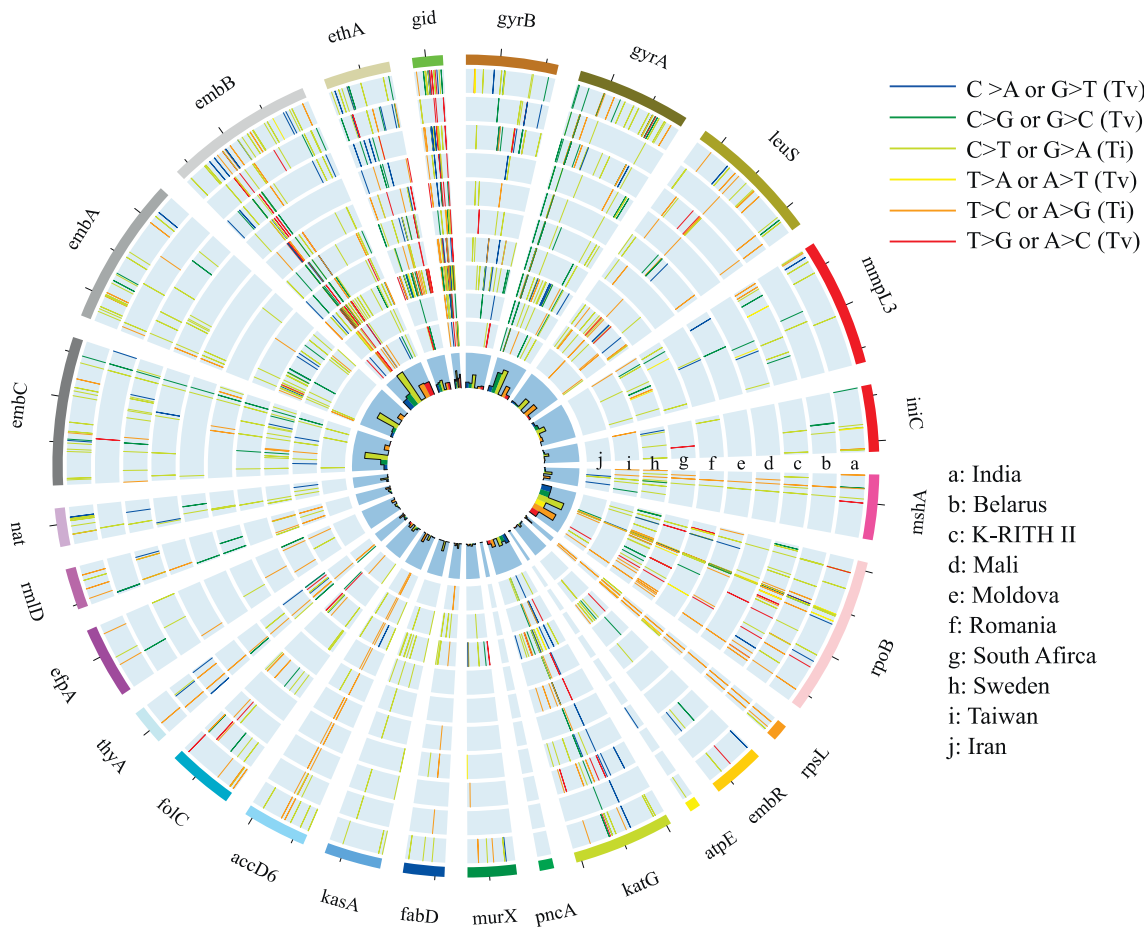


Fig. 5. Transition and Transversion pattern for a set of 26 candidate genes across different geographic types. Note: The color highlighters represent the 6 substitution mutation types: Transitions and Transversions observed in drug resistant genes. The outer circle consists of 26 selected drug resistant gene bands. Innermost circle describes the total count of 6 substitution mutation types observed in all the countries for individual gene.

Table 1
Predominant mutation types observed in the candidate resistance genes.

Antibiotic	Associated genes	Predominant mutation types
Rifampicin	rpoB	C > T or G > A and T > C or A > G
Isoniazid	katG	C > T or G > A, C > A or G > T and T > C or A > G
Pyrazinamide	fabD, accD	C > T or G > A and T > C or A > G
	kasA, nat	C > T or G > A
	pncA	Nil
	embC, embA, embB, embR, iniA	C > T or G > A
Ethionamide	rmlD	T > C or A > G
	ethA	C > T or G > A and C > G or G > C
Streptomycin	rpsL	T > C or A > G
	gidB	C > T or G > A and T > G or A > C
Flouroquinolone	gyrA	C > T or G > A and C > G or G > C
	gyrB	C > T or G > A
Para-aminosalicylic acid	thyA	T > C or A > G
	folC	C > T or G > A and T > C or A > G
Sutezolid	leuS	C > T or G > A and T > C or A > G
Capuramycin	murX	C > T or G > A and T > C or A > G

the requirement. The schematic representation of MutVis workflow is shown in Fig. 1. The Snakemake directed acyclic graph (DAG) of jobs run is shown in supplementary material (Fig. S1). After completion of the analysis, a detailed HTML report is generated on runtime statistics, workflow topology, and output description.

2.1. Generation of VCF files from whole genome raw reads

Most bacteria being haploid organisms, GATK UnifiedGenotyper is recommended for genome analysis in non-diploid organisms (Weldat-sadik et al., 2017; Saltykova et al., 2018; Greig et al., 2019). The workflow installs the required software to perform variant calling on users provided whole genome data in FASTQ format. Initially, quality control and cleaning are performed using PRINSEQ (Schmieder and Edwards, 2011). Read mapping on reference sequence is performed using BWA: BWA-MEM algorithm (Li, 2013) and MarkDuplicates with Picard tools (Institute B, 2016). Realignment and variant calling are performed using GATK, GATK-GVCF mode (Disratthakit et al., 2020). Further, the generated VCF files containing raw calls are filtered with parameters “QD < 2.0”, “DP < 10”, “FS > 60.0”, “MQ < 40.0”, “MQRankSum < -12.5”, “ReadPosRankSum < -8.0”. The INDELS are removed using GATK SelectVariants function and SNPs with PASS filter are retained. The parameters for all the filtering are given in config.yml file and can be changed according to user requirements.

2.2. Transition and transversions frequency and proportion

The processed VCF files are converted to MGI format using a custom script to make it compatible for graphical visualization of transition and transversions types through GenVisR package (Skidmore et al., 2016).

2.3. Generating count matrix

BSgenome data package (Pages et al., 2015) mentioned in MutationalPatterns package (Blokzijl et al., 2018) does not contain all reference genomes to perform the analysis. Hence, we utilized Helmsman in the framework to generate mutation spectra matrices. Helmsman (Carlson et al., 2018) is used for rapid and efficient generation of mutation spectra matrices from large next generation sequencing (NGS) datasets. Helmsman accepts VCF format and reference genome in FASTA

format to generate count matrix spectra. The parameters are set to analyze haploid organisms. Helmsman defines the mutation type for each single nucleotide variation (SNP) observed in a VCF file based on the reference and alternative alleles, then queries the corresponding reference genome for the trinucleotide context of the SNP. If the input VCF file is a combined VCF file and has many sample data, the generated mutation spectra matrix will have counts for all the samples present in the VCF file. To generate a sample type specific matrix representing the entire input VCF file, L2 norm is applied for each 96-trinucleotide mutation value.

2.4. Mutational signature derivation

To make the generated mutation count matrix compatible to the MutationalPatterns package, an R package called musigtools is utilized. The input file is formatted to a tab-delimited format with 96 features (base substitution subclasses) as column names and sample names as row names. This file is provided as input to MutationalPatterns package which utilizes the Non-negative matrix factorization (NMF) (Hutchins et al., 2008) in-built function to generate mutation signatures. An essential parameter in NMF is the factorization rank, which is used to estimate the number of mutational signatures. The most general method is to select the smallest rank for which cophenetic correlation coefficient starts reducing. Another method is to select the rank for which the plot of the residual sum of squares (RSS) between the input matrix and its estimate shows an inflection point (Hutchins et al., 2008; Ji et al., 2015; Gaujoux and Seoighe, 2010; Brunet et al., 2004). The number of mutation signatures must be small compared to the number of samples in the mutation matrix.

3. Results and discussion

MutVis is automated to analyze multiple datasets and provides visualization of mutational signatures in bacterial genomes. This framework provides 3 step analysis once the VCF files are available: (i) Whole genome TvTi plot generation, (ii) Gene level TvTi plot generation and (iii) Mutation signature analysis.

To demonstrate and validate the current framework, the VCF files were downloaded from the PATRIC TB-ARC Antibiotic Resistance Catalog consisting of geographically diverse *Mycobacterium tuberculosis* drug resistant strains from Belarus, India, Iran, K-RITH II (South Africa), Mali, Moldova, Romania, MRC (South Africa), Sweden and Taiwan. The description of the dataset is provided in the supplementary material (Table S1). The VCF files are cleaned to retain only the samples of *Mycobacterium tuberculosis*. The VCF files are combined using VCFcombine from Galaxy web-based platform (Blankenberg et al., 2010) to generate a unified VCF file – one for each of the individual geographical location. For all the SNPs observed in VCF files, the density of SNPs is calculated using a sliding-window size of 5 kb throughout the *Mycobacterium tuberculosis* H37Rv genome and constructed SNP density plot using Circos (Krzywinski et al., 2009) (Fig. 2).

3.1. MutVis-genome mutation TvTi plot generation

For classification of genome-wide SNPs into 6 base-pair substitution mutation types, MutVis-Genome mutation TvTi plot generation can be used to plot proportion and frequency of each transition and transversion type. Results of the analysis of *Mycobacterium tuberculosis* samples with MutVis have shown a larger proportion constituting the transition, with the rate of G > A or C > T and A > G or T > C transitions

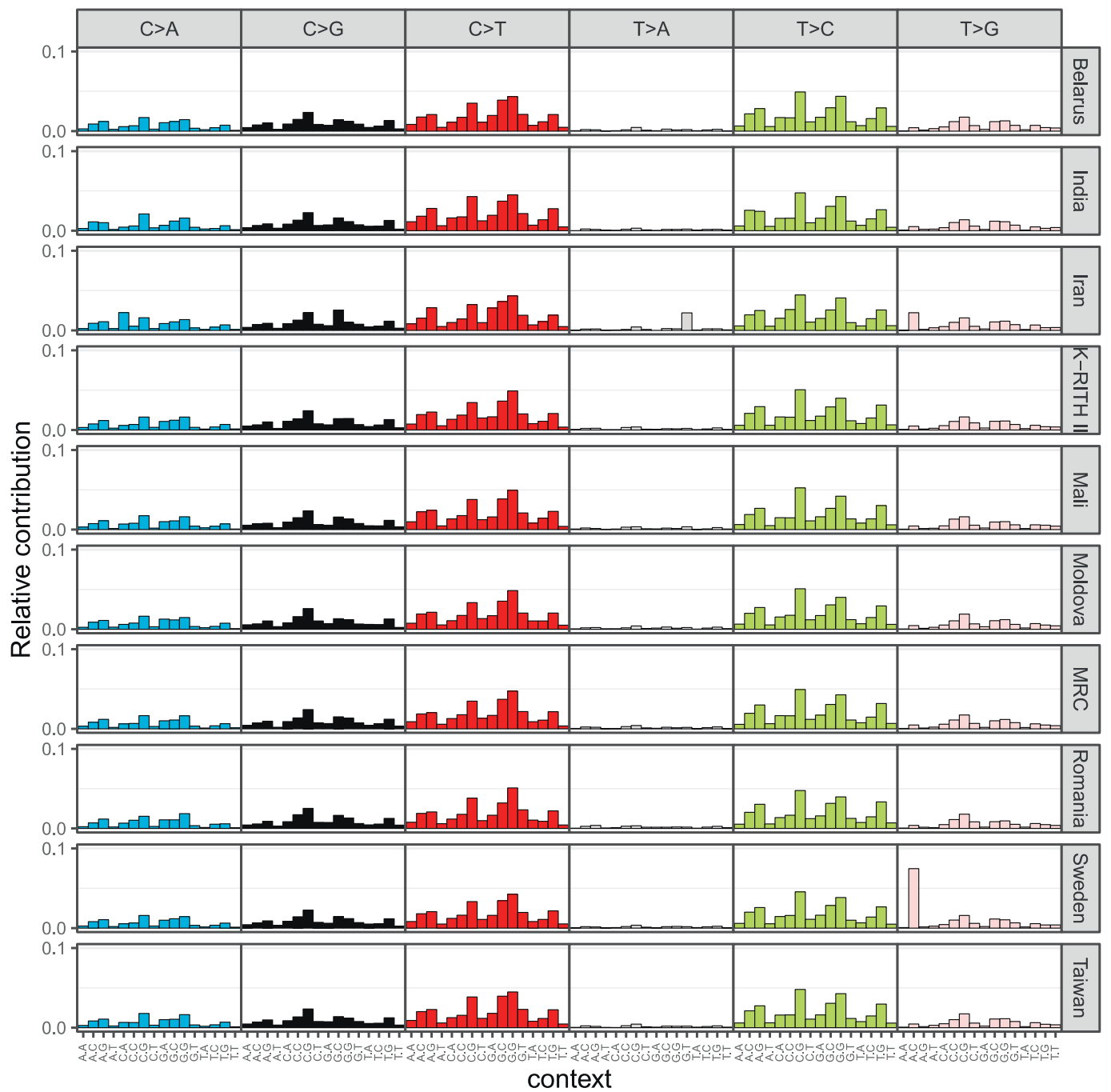


Fig. 6. Genome-wide 96 profile base substitution spectrum.

Note: The probability bars for the six types of substitutions are displayed in different colors. The 96 substitution types are on the horizontal axes, while vertical axes depict the percentage of mutations attributed to a specific mutation type.

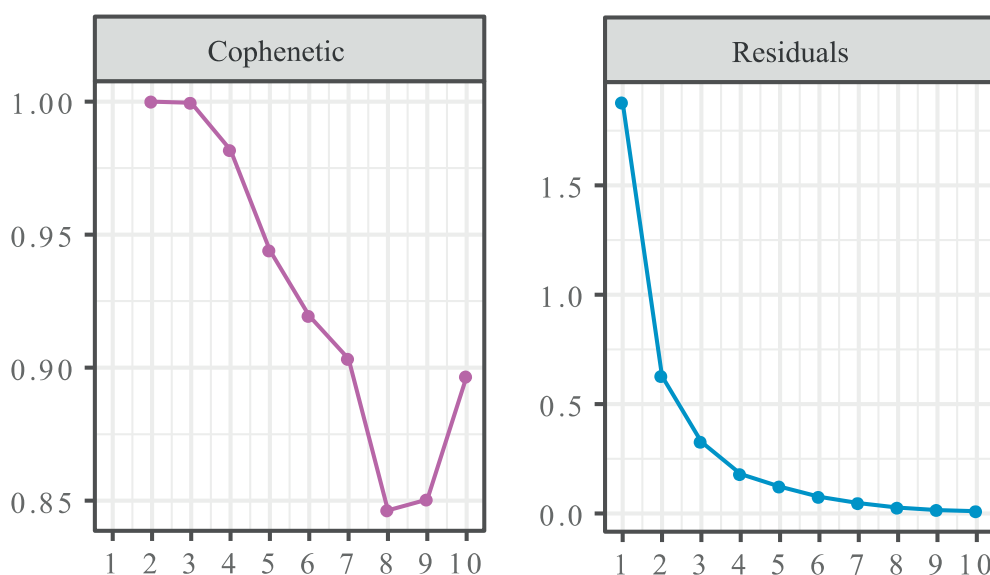


Fig. 7. NMF Rank Survey. X-axis: rank.

exceeding other base pair mutation types (Fig. 3). Among which, G > A or C > T transitions occurred more frequently than A > G or T > C. This observed transition bias in the resistance strains of *Mycobacterium tuberculosis* are in accordance with a similar study (Payne et al., 2019) and therefore validates the framework results.

3.2. MutVis-gene TvTi plot generation

The MutVis framework can also be used to further understand the mutational bias in the individual gene level. The user needs to provide the start and end genomic positions of the required gene in the console. For the current demonstration, we have considered 26 candidate resistance genes important in understanding antibiotic resistance in *Mycobacterium tuberculosis*. MutVis-Gene TvTi plot report for one of the gene-rpoB is provided in the Fig. 4. The 6 base pair substitution pattern analysis for a set of 26 candidate genes is represented in the Circos plot given in Fig. 5. Predominant mutations observed in the investigated genes are tabulated in the Table 1. Identical to the whole genome transition bias, the resistant genes have also shown higher number of transitions predominant in either or both G > A or C > T and A > G or T > C mutation types (Payne et al., 2019).

3.3. MutVis-mutation signature analysis

To obtain further resolution of observed dominant G > A or C > T and A > G or T > C mutation type in terms of sequence context, 96 mutation subtypes classification is analyzed. The resultant base substitution spectrum plot is shown in Fig. 6. For our dataset, genome wide analysis has revealed high mutation in trinucleotides 5'GCC3' > 5'GTC3', 5'GCG3' > 5'GTG3' in the class C > T and 5'CTG3' > 5'CCG3' in the class T > C substitution (J-i et al., 2007; Rosales-Klintz et al., 2012; Madrazo-Moya et al., 2019; Bolotin et al., 2009).

The rpoB gene analysis has shown dominant 5'TCG3' > 5'TTG3' in all samples except in Indian sample which has shown higher 5'TCA3' >

5'TTA3' sequence context for the class C > T; and 5'CTG3' > 5'CCG3' mutation dominance in T > C class (Supplementary Fig. S2). These mutations are reported in the previous study of mutation associated with rpoB gene in drug resistant mycobacterium strains (J-i et al., 2007; Umar et al., 2020; Matsui et al., 2020; San et al., 2018; Song et al., 2016).

Regarding derivation of mutation signatures, user need to provide the number of required signatures on the console based on provided NMF rank survey plot. For the analyzed samples in this study, factorization rank is set to generate two mutational signatures based on Fig. 7. The generated signatures: Signature A and B displaying 96 trinucleotide changes is shown in Fig. 8. The relative contribution of each signature for each geographic type is plotted as a heatmap to display clustering of data based on signatures as shown in Fig. 9. Iran and Sweden samples are mainly characterized by signature A, and the rest of the samples are predominantly characterized by signature B. This will facilitate the classification of the samples with similar mutational signatures. Characterization of mutational signatures contribution at sample level facilitates the identification of the underlying mutational processes.

4. Conclusion

Our study presents the potential of the mutational signature analysis to characterize mutations in drug resistant bacterial genomes. MutVis allows the analysis at multiple samples, which is necessary for future clinical implementation of this framework. Easy implementation and comprehensive analysis of mutation signatures in bacterial genome are key factors of our framework. The potential clinical utility of the present framework includes genetic characterization of bacterial population through BPSs and mutation signature analysis in MMR-defective strains, drug resistant strains and hyper virulent strains.

The MutVis framework is validated on a *Mycobacterium tuberculosis* drug resistant strain dataset. Base substitution profiling revealed transition bias with dominant G > A or C > T and A > G or T > C mutation type at whole genome and in the selected drug resistance genes. The

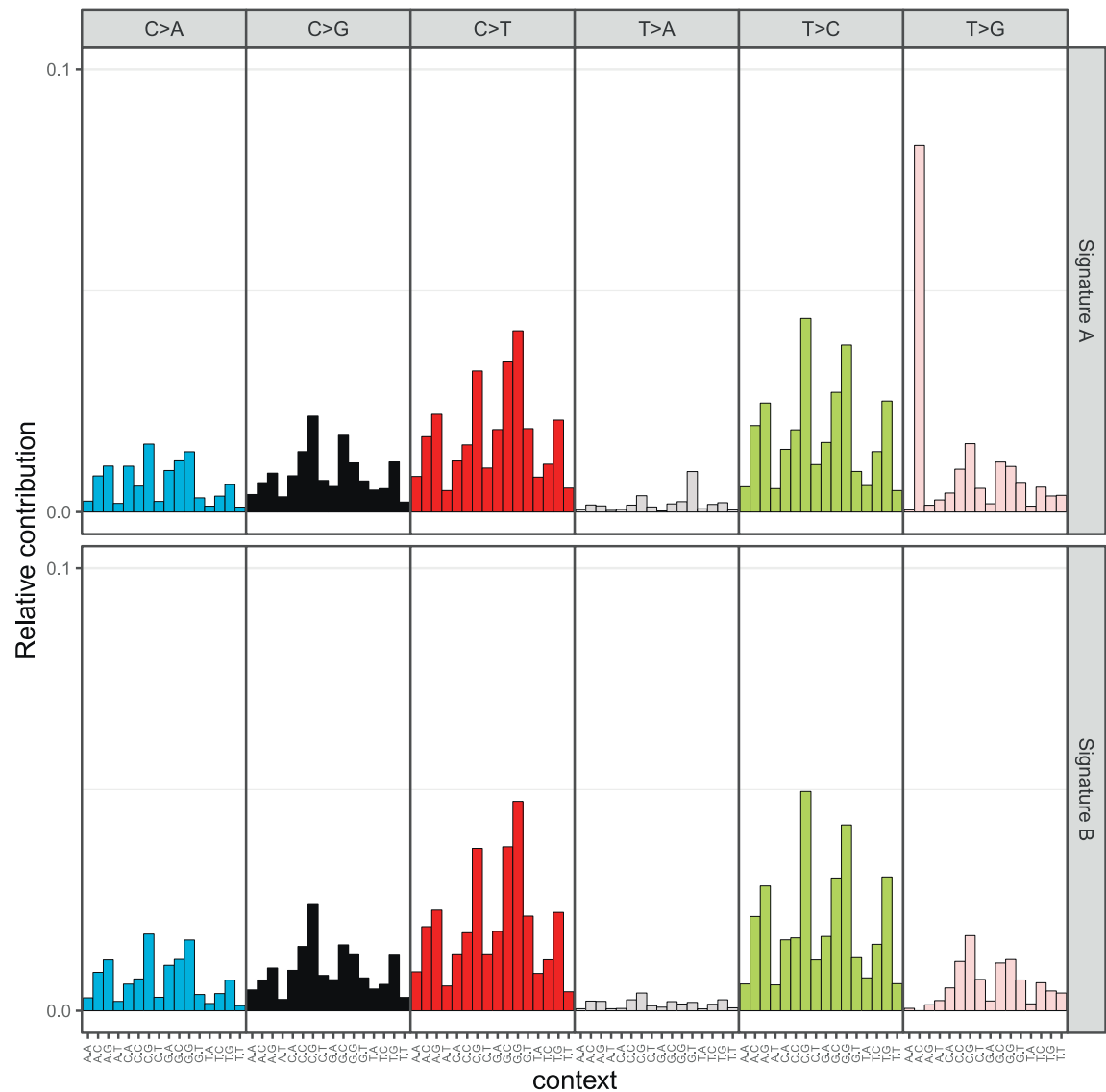


Fig. 8. 96 profile of generated signatures: signature A and B.
Note: The 96 substitution types in the two mutational signatures that were identified by NMF analysis.

mutation signature analysis displayed higher mutations in 5'GCC3' > 5'GTC3', 5'GCG3' > 5'GTG3' and 5'CTG3' > 5'CCG3' sequence contexts for genome wide analysis. The identified mutations are validated in comparison with the earlier studies on *Mycobacterium tuberculosis* drug resistant strains. Two mutational signatures were derived and classified the data into two clusters which is visualized through heatmap profile. In conclusion, the capability of MutVis framework to identify the mutational signatures in clinical samples has the potential to detect mutational bias and mutational signatures at gene and genome level. This will provide insights in diagnosis and treatment strategies of infectious diseases.

Availability and requirements

- Project name: MutVis.
- Project home page: <https://github.com/AkshathaPrasanna/MutVis>
- Operating system(s): Linux.
- Programming environment: Snakemake, Python, R, Bash.
- Other requirements: Python 3.7 or higher, Conda.
- License: MIT.
- Any restrictions to use by non-academics: N / A.

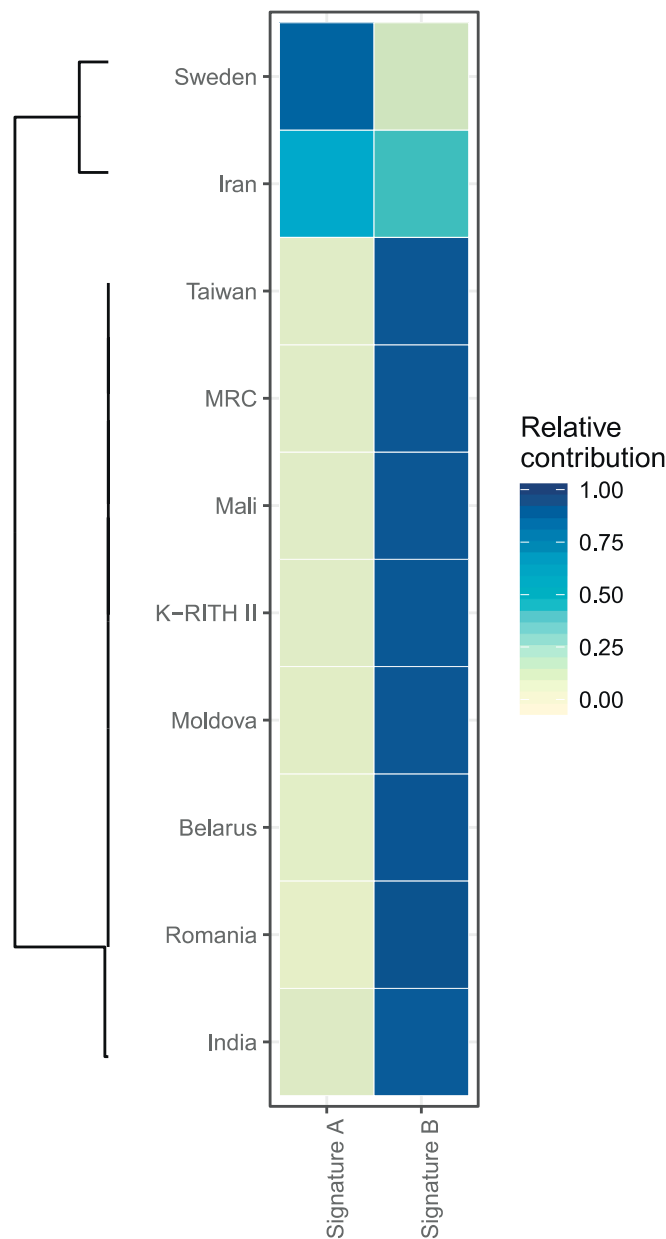


Fig. 9. Heatmap of cosine similarities between the mutational signatures of each sample.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.meegid.2021.104805>.

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